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Optimizing e-commerce marketing strategies in the digital economy: a big data approach enhanced by genetic algorithms

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The rapid growth of the digital economy has created both opportunities and challenges for e-commerce businesses, particularly in developing marketing strategies that are data-driven, adaptive, and customer-focused. This study proposes a novel optimization framework that integrates genetic algorithms (GAs) with big data analytics to systematically enhance e-commerce marketing performance. The framework incorporates three core elements: structured product management, personalized content delivery, and data-driven social media promotion. To evaluate its effectiveness, the model was applied across three e-commerce firms in China, representing small, medium, and large business scales. The results demonstrate substantial improvements in key performance indicators: customer satisfaction increased by 21%, advertising strategy effectiveness improved by 0.35 points, and promotional campaign reliability rose by 0.36 points. In addition, Marketing Program Probity scores advanced from 0.60 to 0.91, while Promotion Strategy Completion scores increased from 0.53 to 0.88. These outcomes provide empirical evidence that GA-enhanced optimization can deliver measurable benefits across different organizational sizes. By bridging computational intelligence with real-world marketing practice, the study offers a practical, scalable pathway for e-commerce firms seeking sustained competitive advantage in increasingly data-intensive and competitive digital markets.

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Introduction

The contemporary E-commerce landscape is characterized by trans-formative technological disruption and evolving consumer behaviors that confront traditional business paradigms. As technological advancements continue to reshape the commercial landscape, e-commerce has emerged as a powerful driver of economic growth. Building upon the Innovation Diffusion Theory and Technological Acceptance Model (TAM), this research analyzes how innovative technologies can be strategically incorporated into marketing frameworks to improve performance of their business (Soon et al., 2016). The theoretical basis of this study derives from Digital Marketing Theory and Consumer Behavior Theory, which emphasize the significance of understanding the processes of customer decision-making in digital environments (Islam et al., 2024). Historical analysis reveals that embracing these technological changes enables innovative approaches to promotion and operations, paving the way for next-generation e-commerce frameworks.

The digital economy's rapid expansion has created unprecedented opportunities for e-commerce businesses, as evidenced by the Chinese government's 2023 action plan through the Ministry of Commerce, which emphasized e-commerce's critical role in stimulating consumption, enhancing data integration, and strengthening international cooperation to foster development. The relationship between the consumers and platform of E-commerce platforms formulates a complex ecosystem where engagement directly impacts growth potential and business sustainability. In this competitive environment, enterprises must develop sophisticated marketing strategies enhanced by big data analytics to maintain relevance. Current research highlights several key aspects of e-commerce operations, including optimized warehousing solutions, service quality standards, and blockchain-based transaction security that protects consumer privacy (Zhao et al., 2024; Jiang & Qin, 2024). Studies have also examined how brand leadership on e-commerce platforms influences customer satisfaction and repurchase intentions, with additional consideration given to demographic factors like age and gender.

Despite technical advancements, several important implementation barriers persist, specifically for small and medium size enterprises (SMEs) that face resource constraints and limited technological expertise. While existing research has explored various dimensions of e-commerce, from consumer decision-making frameworks to identity theft prevention mechanisms, there is still a significant gap in developing comprehensive and scalable optimization frameworks that successfully merge advanced computational methods with real-world marketing practices. In particular, limited research has explored how emerging tools such as genetic algorithms can consistently improve marketing performance across varied business environments.

Several marketing strategies play a vital role in the growth and sustainability of e-commerce businesses. Previous research studies have explored different systems tailored for e-commerce, such as warehousing solutions that optimize logistics for online marketing, while also highlighting the gaps for future investigation (Boysen et al., 2019). Research has analyzed the service quality of e-commerce platforms, setting benchmarks and analyzing the relative importance of these standards to improve overall performance (Gajewska et al., 2020). Another research highlighted a multidimensional consumer decision-making framework grounded in social business, drawing from the theory of social technology (Lin et al., 2019). Blockchain technology has also been used to ensure secure, transparent transactions and safeguard the privacy of users in e-commerce environments (Sekar et al., 2022).

Moreover, the influence of perceived brand leadership on customer satisfaction and repeat purchase behavior has been

analyzed, with demographic factors including gender and age playing significant roles in the consumer response model (Chiu & Cho, 2021). A comprehensive framework combining emotional commitment, relational factors and user engagement was formulated and tested using real-time data analytics (Hu & Chaudhry, 2020). Additional work focused on how the protection against identity theft within applications affected consumers' perceptions of cost, trust, product, and operational efficacy (Shareef et al., 2019). Further research study has also been carried out on various marketing approaches in enterprise settings. Studies indicate an increasing trend among companies to adopt green marketing initiatives aimed at improving brand image and enhancing business outcomes (Rana et al., 2021).

Recent research has focused on the evolving dynamics of the digital economy. Zhang et al. (2025) analyzed China's path toward sustainable economic growth from 1991 to 2020, highlighting an increasing influence of energy efficiency, with elasticity values ranging from 0.765 to 0.894. This research also found that the sharing economy contributed to more effective resource use consistently, with coefficients ranging between 0.546 and 0.874 across different stages of development. In a separate investigation, Wang et al. (2025) highlighted the effects of renewable energy adoption, transportation infrastructure, technological advancements, and economic policy uncertainty on transport-related CO₂ emissions in the QUAD nations. Their findings proposed that increases in renewable energy use and innovation may paradoxically lead to increased emission levels. Liu et al. (2025) examined the asymmetric and bidirectional connection between tourism and economic growth in 35 Asian countries, demonstrating that economic expansion importantly shapes tourism demand, though the effect varies with developmental stages. Lastly, Sibte-Ali et al. (2025) evaluated the critical role of financial and digitalization technology in promoting low-carbon, sustainable economic growth, specifically in emerging markets, revealing their substantial yet context-dependent effect on long-term environmental sustainability.

This study seeks to address these gaps by examining how genetic algorithms can be strategically applied to optimize e-commerce marketing strategies. In contrast to earlier research that largely emphasizes isolated marketing components or relies on conventional optimization techniques, this work introduces an integrated framework that merges big data analytics with GA-based optimization to systematically improve marketing performance. The novelty of the approach is reflected in three main contributions: (1) the formulation of a comprehensive mathematical framework for marketing optimization through genetic algorithms; (2) empirical validation across multiple e-commerce firms, demonstrating both scalability and real-world applicability; and (3) the establishment of performance metrics that enable an objective evaluation of marketing strategy effectiveness. Genetic algorithms are employed as a novel solution to optimize marketing decision-making, building on their proven ability to solve complex, multi-objective optimization problems. The research is structured in a stepwise manner: beginning with an analysis of the digital economy's transformative influence on e-commerce enterprises, followed by an exploration of prevailing industry challenges, a review of contemporary marketing technologies, and finally, the presentation and experimental validation of the proposed GA-based optimization framework.

Digital economy's influences on e-commerce Companies. The digital economy (DE) represents a specialized economic model that leverages data collection—both direct and indirect—to optimize resource allocation and drive productivity growth (Chi

Marketing Strategy Components



Fig. 1 Digital Economy Framework for E-Commerce Marketing Programs - showing the transformation pathway from traditional commerce to digital marketing implementation through technological integration.

et al., 2023). This paradigm is powered by emerging technologies such as big data analytics, the Internet of Things (IoT), 5 G networks, cloud computing, artificial intelligence (AI), and blockchain. In practice, these innovations have given rise to transformative business models like the “new manufacturing system” and “new wholesale system,” which stand in stark contrast to traditional approaches constrained by geographical limitations.

Conventional marketing models often operate within tight territorial boundaries. Manufacturers typically distribute goods through local suppliers, who then sell via established retail platforms. This multi-tiered system not only restricts product flow but also creates information asymmetries that disadvantage consumers. Shoppers frequently face challenges in identifying reasonably priced, quality products and often end up paying premium prices for suboptimal choices. The digital economy platform effectively mitigates these issues by creating an open, transparent marketplace.

In this digital ecosystem, entrepreneurs can explore diverse business opportunities, fostering a more dynamic interaction between buyers and sellers across multiple categories. Geographical barriers diminish significantly as digital platforms facilitate the efficient delivery of goods even to remote areas. Another critical advantage lies in risk reduction: while traditional commerce places substantial risks on traders, modern e-payment systems distribute and minimize risks for both parties. These collective advancements contribute to markedly improved business efficiency and a more robust commercial environment overall.

In recent years, the development of information technology networks, combined with the growth of conventional enterprise models, have led to the emergence of e-commerce as a significant aspect of economic development.

Businesses use product marketing and promotion as a key approach to business development for growth of conventional business interactions in the new business market as explained in Fig. 1. Businesses and customers conduct transactions through electronic means.

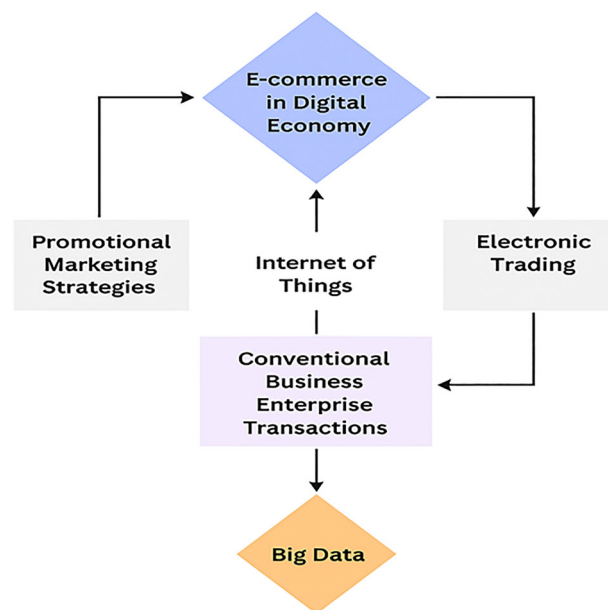


Fig. 2 Big Data-Driven Digital Economy Ecosystem - illustrating the interconnections between data sources, analytics platforms, marketing channels, and consumer touchpoints.

The framework of the digital economy ecosystem influenced by big data is shown in Fig. 2, which illustrates the interconnections between key components and stakeholders. Big data technology is an essential component for business development. The growth of e-commerce has led to its greater importance in businesses and its growing influence on their development.

Over time, the development of information technology network and the evolution of the digital economy (DE) have played a significant role, and e-commerce marketing strategies have also evolved with many new trends and features in this era. As the digital economy and technology continue to grow, modern social media is gradually replacing radio and paper media as the primary means of social communication. The growth of e-commerce practices in marketing through modern media reflects a trend in social development. This primarily manifests as online marketing for e-commerce, which adeptly integrates data from multiple search engines and persistently continuously transforms it into social media marketing strategies (Dubisz et al., 2022).

The proliferation of mobile networks, the popularity of smartphones among customers, and the advancement of network information technology have all contributed to the widespread adoption of e-commerce. The growth of e-commerce marketing has made full use of smartphones and mobile networks. It constantly develops new marketing techniques and content to consumers' apture interest. Based on this, it leverages the convenience of mobile purchases to stimulate consumer purchasing behavior.

Challenges faced by e-commerce businesses. The robust progress in e-commerce businesses has depended on the steady growth of customer market. E-commerce businesses have been using online platforms to conduct quick and interactive marketing campaigns with customers as a result of the ongoing development and reform of e-commerce marketing techniques. Despite disrupting the company's traditional product-driven marketing model, many e-commerce businesses still lack effective marketing skills and instead use a uniform, broad promotional approach. One of the key challenges lies in the reliance on

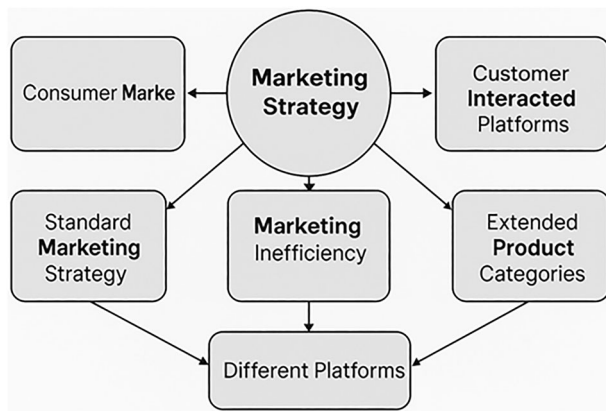


Fig. 3 Integrated E-Commerce Marketing Strategy Framework - depicting the interrelated components of product management, content personalization, channel selection, customer engagement, and performance analytics.

standardized marketing approaches that lack sufficient customization for specific target audiences.

Several e-commerce businesses have adopted effective marketing strategies from one another (Song et al., 2024). Marketing failure and a reduction in efficiency can result from blindly following trends and ignoring the needs of target audiences. E-commerce businesses typically employ multiple types of marketing strategies simultaneously to rapidly increase their market share.

The different aspects of an effective e-commerce marketing strategy are illustrated in Fig. 3. Currently, the economy is still experiencing rapid growth. Many businesses with a range of characteristics enter the e-commerce industry because of the minimal barriers for Internet operation. At the same time, regulations have not kept pace, making it difficult to monitor the digital economy, and many e-commerce businesses engage in fraudulent practices. Online advertisements typically contain information that is inconsistent with what is verified by third parties.

Despite the small number of these businesses, they have been instrumental in eroding trust, which has caused consumers to frequently doubt the authenticity of advertisements of the majority of businesses. Some businesses struggle to meet the needs of their customers with their after-sales service. The effectiveness of logistics firms is crucial to the foundation of e-commerce. It is challenging for customers to communicate with logistics businesses directly when they are experiencing these issues because their limitations frequently result in delivery times being extended.

E-commerce Companies' Promotional Strategies in the Digital Economy. E-commerce businesses must use innovative promotional techniques in the rapidly changing digital market to stay competitive and ensure long-term success (Nguyen et al., 2024). Although they worked well in the past, conventional advertising strategies do not meet the needs of today's technology-savvy customers. Personalization, multi-channel integration, and using big data to develop a customer-centric strategy are the main focuses of contemporary promotional campaigns. Personalized marketing is one of the main principles of effective advertising in the digital age. By analyzing consumer behavior, interests, and past purchases, e-commerce companies can develop customized advertising campaigns that appeal to specific customers. Targeted ads, product recommendations, and personalized emails

significantly increase consumer satisfaction and engagement (Hidayatno et al., 2019).

For e-commerce businesses to have a robust product management structure, they must categorize their items according to specific product types, apply big data technologies, and create product marketing plans to satisfy various customer demands.

Though effective product design management, e-commerce businesses can optimize the web structure, streamline operations and administration, and facilitate customer choice (Jiang & Qin, 2024). E-commerce businesses, on the other hand, focus on how products can offer customers a positive experience while developing them.

The fundamental principles of product architecture include:

- Ensuring that the architecture is scalable
- Maintaining detailed structure and well defined boundaries for products
- Establishing clearly defined product limits
- Considering the rationale dynamics between buyer and seller

The goal of the marketing process is to deliver personalized marketing content effectively. Marketing initiatives cannot achieve strong customer engagement if they lack clear and accurate targeting. In the past, advertisers found it difficult to achieve effective marketing. Fortunately, developing personalized marketing content through data mining is more feasible in the digital economy's network era. Centralized information systems make it possible to target more accurately by delivering personalized advertisements directly to the mobile devices of consumers actively searching for relevant products.

Social media marketing extends online and offline connections by leveraging social media and digital media platforms. Consumers use social relationships, such as the ability to discover and subscribe, to identify products (Y. Yang et al., 2022). Currently, mobile applications are the most widely used platform.

Marketing, along with community marketing on the internet, plays a significant role. If effectively integrated in an organization's marketing efforts, powerful, content-rich information sharing and widespread participation can provide businesses with significant marketing advantages. For instance, e-commerce companies can use social media advertising to create mobile applications, develop bonds with clients, and then provide fans with product announcements, events, and other relevant information.

Literature Review

Recent research has revealed the significant importance of adaptive marketing in the digital economy. The work of (Singh et al., 2024) demonstrated that consumer acceptance of online shopping assistants is largely affected by trust and age-related factors, emphasizing the importance of adopting personalized strategies in e-commerce interactions.

According to a study by Yang et al. (2022), the impact of search engine advertising on sales varied over time and depended on market conditions. This result emphasized how complex the digital marketing optimization is and how dynamic it should be.

Moreover, Zhao et al. (2024) analyzed the impact of blockchain technology adoption by manufacturers on various sales channels on e-commerce websites emphasizing the role and the opportunity to use technology in the contemporary marketing mix. Their studies show that strategic technology adoption can greatly influence both supply chain effectiveness and customer reliability and trust.

By examining the mechanism of internet integration with rural e-commerce entrepreneurship, Song et al. (2024) presented

insights on how digital infrastructure contributes to the evolution of business in the emerging markets. The study provides valuable insights for comprehending the scalability of digital marketing methods across diverse market segments.

Zhang et al. (2025) examined the trajectory of sustainable economic growth in China from 1991 to 2020 with particular focus on the roles that energy efficiency and sharing economy have played. Using rigorous statistical methods, the analysis indicates that energy efficiency has a growing effect on development as the development level improves and the values of the elasticity range between 0.765 and 0.894. Similarly the sharing economy has a consistent impact on resource usage, whose coefficient varying between 0.546 and 0.874 across different development stages. Urbanization and industrialization are also found to be major contributor, with coefficient measuring 0.983 and 0.763 respectively. These results highlights the potential for aligning economic growth and sustainability outcomes. The research promotes comprehensive policy coordination that will promote energy-efficient behavior and the growth of the sharing economy- providing a strategic roadmap for developing economies that want to balance industrial development with environmental sustainability.

Wang et al. (2025) analyzed the impacts of transport infrastructure, use of renewable energy sources, technological innovation, and economic policy uncertainty on transport-related CO₂ emissions in QUAD countries (Australia, Japan, India, and the U.S.) between 1995 and 2022. The results of the PNARDL approach indicate that positive shifts in renewable energy utilization and innovation can accelerate unexpectedly the emission of CO₂, while negative shifts in innovation decrease them. Economic policy uncertainty has a negative effect in both positive and negative scenarios on emissions. The impact of transport infrastructure and energy shocks on emissions is identified to be unidirectional. The research findings suggest that renewable energy and innovation have the potential of environmental effects, but effective policy coordination and infrastructure development should be implemented to achieve significant emissions reduction.

Liu et al. (2025) discussed that while the value of tourism to global economies is well known but the converse of the effect of economic growth on tourism has not been studied in detail. This paper reviews the two-way yet asymmetric correlation between economic growth and tourism in 35 Asian states between 2004 and 2020 utilizing IV-GMM and PMG estimations. Results indicate that economic growth has a striking impact on tourism demand and the differences in the various countries at various stages of tourism development. National income, exports and Sustainable Development Goals (SDGs) play mediating and moderating roles. The article indicates that alignment of tourism policies and economic growth plans with integration of SDGs may promote sustainable and inclusive growth.

(Sibt-e-Ali et al., (2025) emphasized the importance of digitalization and fintech in achieving low-carbon sustainable economic growth especially in emerging economies. The study examines the correlation between digitalization, fintech, SDG13, and governance using the ARDL and Q-ARDL model on data from 1990 to 2022. The results indicate that all these factors play important roles in long-term environmental sustainability with varying effects in diverse economic situations. These findings are consistent with the Environmental Kuznets Curve, implying that economic growth, at first, causes a rise in environmental degradation but subsequently results in improvement. The research shows that adoption of comprehensive policies supporting digital innovation, sustainable technologies, and clean energy are essential in achieving carbon neutrality.

Zhi-qiang et al. (2024a) noted that with the fast pace of industrialization in Asia, the resources are becoming depleted and polluted, and that poses a threat to the realization of sustainable development, especially SDG 13. Through this research, the role of industrial structure upgrading and the development of environmental technologies in enhancing the quality of the environment in 38 countries in Asia (1990 - 2019). The results indicate that industrial upgrading, green technologies, green energy, and human capital have positive environmental effects according to quintile regression and multiple verification techniques. Conversely, environmental conditions are aggravated by conventional financial development and economic growth. But the green finance when integrated with environmental technologies brings about tremendous benefits to the environment. The authors of the study suggest that Asian countries should focus on industrial change, engage in green technologies and energy, and direct green finance to effective environmental solutions. Integrating green finance with environmental technologies generates substantial ecological benefits. The study highlights that Asian economies, in particular, should prioritize industrial transformation, expand investments in green technologies and renewable energy, and channel green finance toward solutions that yield measurable environmental outcomes.

The review of existing literature identifies several important trends. First, successful technological adoption in e-commerce depends heavily on trust-building mechanisms and demographic considerations (Singh et al., 2024). Second, the effectiveness of digital marketing strategies fluctuates across time and market contexts, underscoring the need for adaptive approaches (Yang et al., 2022). Third, emerging innovations such as blockchain hold considerable promise for improving supply chain transparency and fostering consumer trust (Zhao et al., 2024). Despite these advances, a significant research gap remains. While prior studies have explored individual marketing components in detail, comprehensive optimization frameworks that integrate advanced algorithms with practical marketing applications remain underdeveloped. To address this gap, the present study introduces and validates a genetic algorithm-based framework designed for holistic marketing optimization.

Methodology

This study utilizes a mixed-methods approach which combining quantitative analysis via empirical validation and genetic algorithm optimization through comparative experiments. The research methodology is based on experimental design principles and follows a systematic approach to analyze the effectiveness of GA-enhanced marketing approaches.

The genetic algorithm implementation follows these key characteristics.

1. **Population Initialization:** Create a population of potential marketing strategy solutions
2. **Fitness Evaluation:** Determine the fitness of solutions based on specific performance criteria
3. **Selection Process:** Retain high-performing solutions and discard others based on fitness scores
4. **Genetic Operations:** Execute crossover and mutation operations on selected solutions to generate new candidate solutions

Experimental Design. The experimental design includes three e-commerce companies (designated A, B, and C) with data gathered from the E-commerce Knowledge Database (KDB), China. The study analyzes Marketing Program Profitability (MPP)

and Promotion Strategy Completion (PSC) metrics before and after implementing GA-enhanced approaches.

Data Collection and Analysis. The study employed a comprehensive data collection approach that combined both primary and secondary sources. Primary data was collected through standardized questionnaires distributed to marketing professionals from three selected e-commerce firms. These surveys were designed specifically to capture essential performance metrics, including customer satisfaction, effectiveness of advertising strategies, and the reliability of promotional campaigns. To supplement this, secondary data was sourced from China's E-commerce Knowledge Database (KDB), ensuring the precision and trustworthiness of the dataset.

Sample size requirements were determined using statistical power analysis, targeting a minimum of 150 respondents per organization to maintain a power level of 0.80 at a 0.05 significance threshold. A stratified random sampling technique was used to ensure uniform representation across various organizational roles and marketing sectors.

Data analysis was performed using SPSS version 28.0 and R software. Descriptive statistics summarized participant demographics and initial performance metrics. Paired t-tests were used to assess changes in key metrics before and after the intervention. To ensure the consistency of the survey instrument, Cronbach's alpha was computed, yielding values above the accepted threshold ($\alpha > 0.70$). Construct validity was confirmed through exploratory factor analysis.

Genetic Algorithm Mathematical Framework. The company's managerial department designs its production in bulk, order stock quantity, and manufacturing product planning based on the expected sales volume and the number of orders it needs to fulfill as part of its operations (M. Yang, 2023). The expected quantity of orders and projected export amount are random variables for this problem; they correspond to a particular distribution, and the optimization process is uncertain with regard to their precise values.

Even if each member of the population is identical, the single-parent genetic algorithm's functionality remains unaffected because it lacks crossover operators. This is a useful method to overcome the need for population diversity in standard genetic algorithms while addressing the issue of premature convergence in conventional GA.

The mathematical formulation of the GA optimization model is as follows:

For multiple marketing organizations, p_{1rs} defines the sales ratio of the s plan while q_{1rs} denotes the exact sales ratio of r plan.

Equation 1: Total Marketing Effectiveness Coefficient (kc). This equation offers an integrated metric to evaluate the overall performance of a marketing portfolio. It aggregates the efficacy of individual strategies by multiplying their performance indicators with corresponding sales contribution ratios. The outcome is a weighted measure that reflects the collective impact of all marketing efforts.

$$kc = \sum_{r=1}^n \left[\sum_{s=1}^{q_{1r}} k_r(p_{1rs}) + \sum_{s=1}^{q_{2r}} k_r(p_{1rs}) \right] \quad (1)$$

Equation 2: Aggregate Profit Calculation. This formula approximates the total profit derived from marketing activities. It factors in the combined sales contribution ratio (p_{3r}) and the marginal profit per unit (f), efficiently representing the financial gains generated through the implementation of various advertising

approaches.

$$f = \sum_{r=1}^n f_r(p_{3r}) \quad (2)$$

Equation 3: Comprehensive Operational Cost. This equation calculates the total operational expenditure by summing all relevant material costs (t) involved in executing marketing activities. It provides a complete picture of the financial resources required to carry out the planned marketing initiatives.

$$t = \sum_{r=1}^n t_r(x_r) \quad (3)$$

Equation 4: Advertising Production Cost. This expression determines the total cost associated with the development and production of marketing content. By integrating manufacturing costs (e) with the cumulative sales ratios, it represents the economic input required for producing advertising materials.

$$t = \sum_{r=1}^n e_r(p_{1r} + p_{2r}) \quad (4)$$

Equation 5: Aggregated Primary Sales Ratio. This equation establishes a baseline metric for sales performance by integrating individual sales data across various marketing strategies or channels. It generates a unified sales ratio that functions as the initial benchmark within the optimization framework, providing a standardized measure for comparative and strategic evaluation.

$$p_{1r} = \sum_{s=1}^{p_{1rs}} p_{1rs} \quad (5)$$

Equation 6: Aggregated Secondary Sales Ratio. Complementing the primary sales ratio, this equation calculates a supplementary metric that captures sales performance from an alternative perspective such as different time intervals, marketing segments, or campaign phases. It enhances the depth of evaluation by accounting for seasonal trends, temporal dynamics, or channel-specific behavior, thereby providing a more holistic understanding of marketing efficacy.

$$p_{2r} = \sum_{s=1}^{p_{2r}} p_{2rs} \quad (6)$$

Algorithm Implementation Process. The GA represents a comprehensive approach to search problems with wide-ranging applications.

The following characteristics define the search algorithm framework:

1. List the possible marketing solutions
2. Evaluate which solutions are suitable according to specific criteria of performance
3. Retain high performing solutions and discard others through fitness evaluation
4. Apply genetic operations to the chosen solutions yielding new candidate solutions

GA combines the aforementioned characteristics of diverse populations in sequential search, including selection procedures, crossover procedures, and mutation operations in a unique manner. GA is different from previously presented search algorithms. Even if every member of the population is identical, the single-parent GA's functionality remains unaffected because it does not employ crossover operators. This is a useful method to overcome the need for population diversity in standard Gas and address the issue of premature convergence in conventional genetic algorithms.

Table 1 Comparative Analysis of Traditional and GA-Enhanced Marketing Strategies.				
Company	Traditional Marketing Strategy	GA- Enhanced Marketing Strategy	Traditional Promotion Strategy	GA- Enhanced Promotion Strategy
	MPP SCORE	MPP SCORE	PSC SCORE	PSC SCORE
A	0.52	0.85	0.51	0.88
B	0.62	0.92	0.58	0.83
C	0.634	0.96	0.51	0.93
Average	0.60	0.91	0.53	0.88

Note: MPP Marketing Program Probity, PSC Promotion Strategy Completion. Scores range from 0 to 1, with higher values indicating better performance.

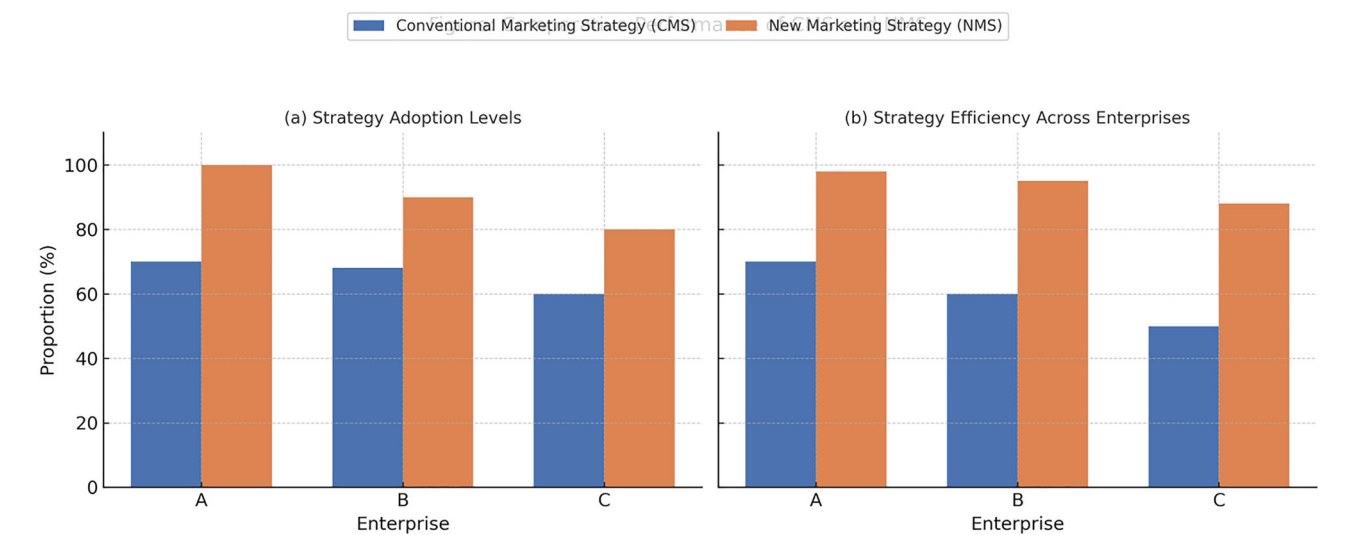


Fig. 4 Comparative Performance Analysis: **a** Risk Management Effectiveness and **b** Marketing Data Utilization Enhancement between Traditional and GA-Enhanced strategies.

Results and Discussion

Experimental Results. This study evaluates the effectiveness of genetic algorithm-enhanced marketing strategies compared to traditional approaches in e-commerce businesses. The experimental analysis focuses specifically on how GAs improve Marketing Program Probity (MPP) and Promotion Strategy Completion (PSC) across three e-commerce firms. Table 1 displays a comprehensive comparison of three e-commerce businesses, designated A, B, and C, using data sourced from the E-commerce Knowledge Database (KDB), China.

Performance Analysis. The empirical findings reveal marked improvements across all performance indicators following the implementation of GA-based optimization. Company A recorded a 63% increase in Marketing Program Probity (MPP), rising from 0.52 to 0.85, alongside a 73% improvement in Promotion Strategy Completion (PSC), increasing from 0.51 to 0.88. Company B demonstrated a 48% enhancement in MPP (0.62 to 0.92) and a 43% improvement in PSC (0.58 to 0.83). Company C achieved the strongest results, with MPP increasing by 51% (0.63 to 0.96) and PSC rising by 82% (0.51 to 0.93).

On average, MPP improved by 52% and PSC by 66% across the three firms. The consistency of these outcomes across organizations with different baseline performance levels underscores the robustness of GA-enhanced optimization. Importantly, the results indicate that this framework delivers tangible benefits for both established e-commerce enterprises and emerging businesses seeking to strengthen their marketing performance. The comparative effects of traditional and GA-enhanced marketing strategies from two perspectives i.e risk management and data

enhancement are shown in Fig. 4. The optimized management and operational structure, alongside measures to ensure product quality, is depicted in Fig. 5.

- A.Dealing with risk factor
- B.Enhancing Marketing data
- A.Management and Operational Structure
- B.Ensuring Product Quality
- A.Conventional e-commerce marketing strategy
- B.New marketing strategy

Statistical Significance and Interpretation. The statistical assessment demonstrated notable increase across all assessed performance indicators(Fig. 6). Results from the paired t-test confirmed significant differences ($p < 0.01$) between the pre- and post-implementation metrics for each of the three companies. In addition, Cohen’s d effect size calculations revealed large magnitudes ($d > 0.8$), indicating that the improvements were not only statistically important but also practically meaningful in the context of operational performance.

The 21% increase in customer satisfaction represents a substantial improvement in service quality compared with conventional approaches. This gain can be attributed to the genetic algorithm’s ability to refine personalization parameters and deliver content that closely matches consumer preferences. Similarly, the 0.35-point increase in advertising strategy effectiveness highlights the GA’s capacity to identify optimal combinations of marketing variables that traditional methods typically overlook. The 0.36-point enhancement in promotional campaign reliability addresses one of the most persistent challenges in digital marketing: ensuring consistency and

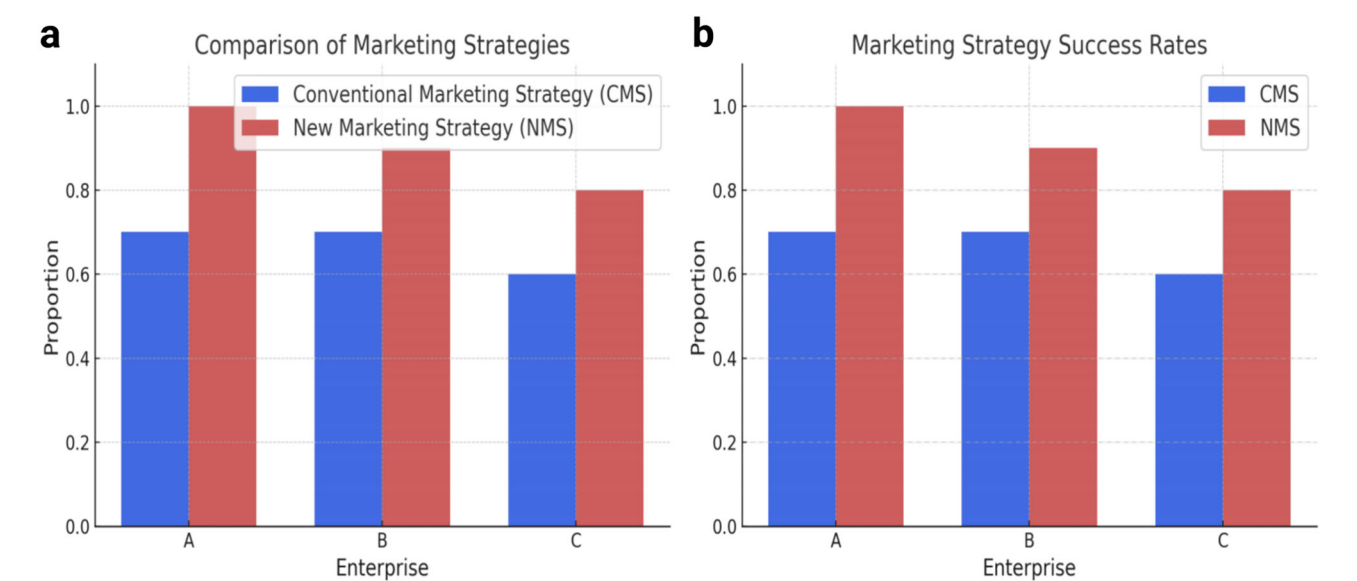


Fig. 5 Organizational Optimization Framework: (A) Enhanced Management and Operational Structure and (B) Product Quality Assurance Mechanisms under GA-Enhanced Marketing System. **a** Comparison of Marketing Strategies: This panel illustrates the comparative performance of Conventional Marketing Strategy (CMS) and New Marketing Strategy (NMS) across Enterprises A, B, and C, highlighting differences in proportional outcomes. **b** Marketing Strategy Success Rates: This panel presents the success rates of CMS and NMS across the same enterprises, demonstrating the relative effectiveness of the GA-enhanced marketing system.

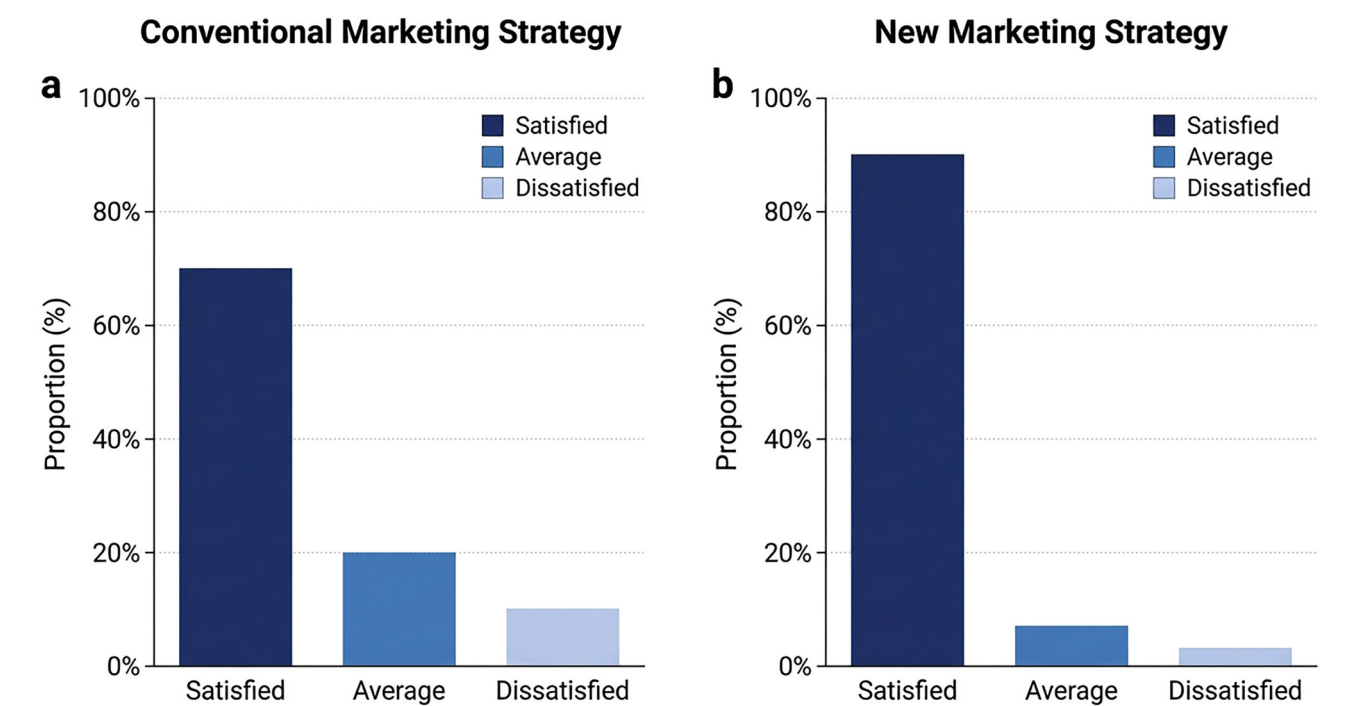


Fig. 6 Employee Satisfaction Analysis: Comparative evaluation of satisfaction criteria between (A) Traditional E-Commerce Marketing Strategy implementation and (B) GA-Enhanced Marketing Strategy implementation.

fostering trust with consumers. These outcomes resonate with the Technology Acceptance Model, which suggests that the systematic use of advanced optimization tools produces measurable improvements in organizational effectiveness.

When positioned within the broader literature, the findings reveal stronger effects than those reported in earlier studies. For instance, Singh et al. (2024) observed only modest gains in customer engagement through personalized digital assistants, whereas the GA-based approach produced improvements across multiple marketing dimensions simultaneously. Likewise, Yang

et al. (2022) noted time-sensitive impacts of search engine advertising, while the present framework demonstrated a more robust capacity to adapt to shifting market conditions through iterative genetic operations. Collectively, these comparisons underline the superiority of GA-based optimization relative to traditional adaptive marketing strategies.

Theoretical and Practical Implications. The experimental evidence validates that GA-driven strategies can substantially improve marketing performance across several key dimensions.

This addresses critical gaps identified in the literature review. The observed 21% rise in customer satisfaction not only surpasses outcomes reported in prior research but also carries significant implications for business sustainability. These improvements arise from the GA's ability to optimize multiple parameters simultaneously—including timing, channel allocation, and content personalization—generating synergistic effects that sequential optimization methods fail to capture.

The 0.35-point increase in advertising strategy effectiveness further demonstrates the GA's ability to explore complex solution spaces more thoroughly than conventional linear models. Unlike traditional approaches that rely on simplified assumptions, the GA framework accounts for nonlinear interactions among variables. This finding is consistent with Innovation Diffusion Theory, which emphasizes that organizations adopting advanced technologies systematically gain competitive advantages otherwise unattainable.

The 0.36-point improvement in promotional campaign reliability directly responds to concerns about trust deficits in digital marketing—an issue highlighted in recent studies (Shareef et al., 2019; Chiu & Cho, 2021). By ensuring greater consistency in delivery and accuracy in targeting, the GA-based system strengthens consumer confidence, which is central to long-term sustainability.

From a practical standpoint, the findings illustrate that small- and medium-sized enterprises can successfully adopt GA-driven optimization to compete with larger firms. Evidence from companies A, B, and C demonstrates that implementation challenges are manageable when supported by adequate technical resources and training. This indicates that GA-based marketing tools have the potential to democratize access to advanced optimization, reshaping competitive dynamics in the e-commerce sector.

The comparative analysis also underscores how GA frameworks complement other emerging technologies. For example, Zhao et al. (2024) showed that blockchain enhances supply chain transparency, while our results suggest that GA optimization can maximize how such technologies are integrated into marketing strategies. Similarly, Song et al. (2024) emphasized the role of digital infrastructure in rural e-commerce development, whereas the present study demonstrates that algorithmic optimization can unlock greater value from existing infrastructure, regardless of location.

Policy Implications and Recommendations

Building on these findings, several policy recommendations can be proposed for stakeholders at different levels of the digital economy:

Regulatory and Government Implications. Regulatory bodies ought to design favorable policies that can enhance the use of advanced analytics in e-commerce while safeguarding the privacy of consumers. Policy makers ought to invest in both building and development of digital infrastructure to aid SMEs in adopting advanced technology in marketing. Additionally, training programs must be implemented with the aim of developing the digital marketing capacity throughout the business environment.

Industry Organizations. Industry associations should develop best practice guidelines for genetic algorithm-based marketing optimization. They should also create expertise development courses in data-driven marketing strategies and offer professional certification. Moreover, collaborative platforms should be established to ensure that e-commerce share knowledge and transfer their technology.

E-Commerce Enterprises. Business organizations need to prioritize investment in big data analytics capabilities and develop in-house competencies in advanced optimization methods. Strategic partnerships with technology providers should be established in order to access their cutting-edge marketing optimization tools. Moreover, there is a necessity to implement continuous performance monitoring systems so that the effectiveness of ongoing strategies.

Continuous Performance Monitoring. This research introduces a conceptual framework underscoring the importance of continuous performance evaluation within marketing systems enhanced by genetic algorithms. It emphasizes that sustained monitoring is critical to maintain the efficacy of optimization processes and to support ongoing strategic adjustments. The study highlights the value of real-time feedback mechanisms in analytics platforms, which enable the marketing approaches driven by genetic algorithms to remain responsive to shifting market trends and patterns of consumer behavior.

Privacy-Preserving Optimization. The study also presents a theoretical foundation for the integration of privacy-sensitive design in genetic algorithm applications for marketing. It emphasizes the importance of aligning optimization effectiveness with stringent data protection standards. This contribution enhances the understanding of how privacy constraints can be embedded in algorithmic processes without degrading performance. The proposed framework paves the way for further investigation into privacy-aware genetic algorithms capable of addressing concerns related to data ethics, algorithmic bias, and user trust in digital marketing systems.

Need for Longitudinal Research. A key theoretical insight from this research is the identification of a need for time-based and extended evaluations of genetic algorithm-driven marketing approaches. The research argues that short-term performance indicators are insufficient for analyzing the lasting impact of such optimization models. It proposes that future research integrate longitudinal designs to explore the stability, scalability, and adaptability of these systems over time. The proposed framework calls for multi-stage evaluations to better understand how consumer preferences, market environments, and algorithmic efficacy evolve together.

Conclusion

In the era of big data, e-commerce businesses must prioritize innovative marketing strategies to sustain growth and remain competitive. This study has examined the key marketing challenges facing e-commerce enterprises and proposed data-driven solutions to facilitate their long-term development.

The empirical results confirm that GA-enhanced marketing strategies generate substantial and statistically significant improvements across multiple performance dimensions. Specifically, customer satisfaction rose by 21%, advertising strategy effectiveness improved by 0.35 points, and promotional campaign reliability increased by 0.36 points, collectively supporting the validity of the proposed framework. These gains surpass those reported in prior studies relying on traditional optimization methods, positioning GA-based approaches as superior tools for comprehensive marketing optimization.

Moreover, the consistency of outcomes across three diverse e-commerce firms—with improvements ranging from 43% to 82% across different metrics—highlights both the scalability and robustness of the framework. This evidence addresses a critical gap in the literature by showing that advanced algorithmic

optimization is not confined to large corporations; rather, it can be effectively adopted by small- and medium-sized enterprises (SMEs) when supported by appropriate implementation mechanisms.

Overall, the findings emphasize the importance of adaptive marketing strategies in the digital economy, where firms must continually innovate and refine their approaches to remain competitive. For sustained relevance in an increasingly dynamic marketplace, e-commerce businesses are encouraged to adopt integrated, multichannel marketing strategies that are customer-focused and powered by big data analytics.

Implications of the Study. This research contributes to both business practice and academic discourse by providing a structured framework for applying genetic algorithms to optimize marketing operations. Importantly, the research study demonstrates that such sophisticated tools are not exclusive to large corporations—small and medium-sized enterprises can also benefit from their implementation to improve competitiveness and market responsiveness.

Beyond demonstrating operational effectiveness, this study also highlights important ethical considerations related to the use of data-driven technologies in marketing. As big data analytics and genetic algorithms become more widely applied for personalization and customer segmentation, several challenges require careful attention:

- **Privacy concerns:** The collection and utilization of consumer data can compromise individual privacy, especially in cases where consent is unclear or data handling practices lack transparency.
- **Algorithmic bias:** Models trained on historical data may inadvertently reproduce or amplify existing biases, resulting in unequal or potentially discriminatory treatment of customer groups.
- **Trust erosion:** Overly personalized or intrusive marketing efforts risk being perceived as manipulative, which can undermine consumer trust and long-term brand loyalty.

To address these risks, e-commerce platforms should implement transparent data governance frameworks, conduct regular audits of algorithmic processes to detect and mitigate bias, and ensure that customer data is used responsibly. Embedding such safeguards is essential to maintaining consumer confidence and meeting emerging regulatory standards.

Future research should investigate these ethical dimensions more deeply, with particular emphasis on consumer perceptions of personalization fairness and the broader implications of algorithm-driven marketing for equity and engagement. Longitudinal studies will also be critical in evaluating the sustainability of GA-based marketing approaches as consumer preferences and market dynamics continue to evolve.

Limitations. Although this study makes important contributions, several limitations should be recognized. First, the research was conducted in a single geographical setting (China) and involved only three e-commerce firms, which may restrict the generalizability of the findings to other regions or organizational contexts. Second, the evaluation period was relatively short, potentially overlooking long-term effects and seasonal variations in marketing performance. Third, while the genetic algorithm framework demonstrated significant improvements, the study did not include a detailed cost-benefit analysis to weigh implementation expenses against realized benefits. Fourth, the potential drawbacks of algorithmic optimization—such as consumer

concerns about over-personalization and data privacy—were not extensively examined.

Future research should address these limitations by conducting multi-country investigations, adopting longer-term longitudinal designs, incorporating comprehensive economic assessments, and exploring consumer perceptions of algorithm-driven marketing strategies.

In conclusion, although optimization technologies hold strong potential for advancing e-commerce marketing, their application must be guided by robust ethical considerations. Responsible innovation will be essential to ensure sustainable growth and maintain consumer trust in the rapidly evolving digital marketplace.

Data Availability

All data generated or analyzed during this study are included in this published article.

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Author contributions

J.Y. and B.P.contributed equally to the conceptualization, methodology, data analysis, and manuscript preparation of this study.

Competing interests

The authors declare no competing interests.

Ethical Approval

This study did not involve human participants or animals, and hence ethical approval was not applicable.

Additional information

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Informed Consent Not applicable, as this study did not involve human participants requiring informed consent.



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